MILITARY SPECIFICATION

TEST SET GROUP, MAGNETIC DETECTOR HEAD, OA-7366/ASM-3, OA-7366A/ASM-3, AND OA-7366B/ASM-3

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1 1 Scope - The test equipment covered by this specification shall be for shop use and shall provide a simulated earth's magnetic field, shielded from the actual earth's magnetic field and all other fixed and/or variable magnetic fields, to the degree specified herein, for calibration of the Detecting Set, Magnetic, AN/ASQ-8 and AN/ASQ-10.

1 2 Classification - The Test Set Group, Magnetic Detector Head, shall be satisfactory for use with the Test Set, Detecting Set, AN/ASM-3, and Test Set, Magnetic Detector, AN/ASM-6(*) The various units which compose the Test Set Group, Magnetic Detector Head, are listed in paragraphs 1.2 1, 1 2 2, and 1 2.3 below

1 2.1 Units - Test Set Group, Magnetic Detector Head, OA-7366/ASM-3, is composed of the following

Items	Type Designation	Applicable Paragraph
Shield, Test	MX-6797/ASM-3	3.4.1
Dolly, Test Equipment	V-341/ASM-3	3.4.3
Simulator, Magnetic	SM-417/ASM-3	3.4.5
Control Simulator	C-6553/ASM-3	3 4.6
Cable Assembly Special Purpose, Electrical	W-1 of OA-7366/ASM-3	3 4 7

1 2.2 Units - Test Set Group, Magnetic Detector Head, OA-7366A/ASM-3 is composed of the following

Items	Type Designation	Applicable Paragraph
Shield, Test	MX-6797/ASM-3	3.4.1
Dolly, Test Equipment	V-341A/ASM-3	3 4.4
Simulator, Magnetic	SM-417/ASM-3	3 4.5
Control, Simulator	C-6553/ASM-3	3.4.6
Cable Assembly Special Purpose, Electrical	W-1 of OA-7366/ASM-3	3.4 7

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1.2.3 Units - Test Set Group, Magnetic Detector Head, OA-7366B/ASM-3, is composed of the following:

Items	Type Designation	Applicable Paragraph
Shield, Test	MX-6797A/ASM-3	3 4 2
Dolly, Test Equipment	V-341A/ASM-3	3 4 4
Simulator, Magnetic	SM-417/ASM-3	3 4 5
Control Simulator	C-6553/ASM-3	3.4 6
Cable Assembly, Special Purpose, Electrical	W-1 of OA-7366/ASM-3	3 4 7

2 APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein

SPECIFICATIONS

Military

MIL-B-18	Batteries, Dry
MIL-T-5422	Testing Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures, Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

DRAWINGS

BUWEPS E-1801	Test for Presence of Magnetic Material

2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot Code DCS, 5801 Tabor Avenue Philadelphia 20 Pennsylvania Drawings may be obtained from the Chief Bureau of Naval Weapons

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and materials requirements shall conform to specification MIL-T-21200.

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(2) Non-repairable subassemblies, as outlined in specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. A mean-time-to-failure of more than 5000 hours should be the goal of each

(3) When previously produced models of this equipment did not use non-repairable subassemblies the design shall not be changed to employ non-repairable assemblies without the approval of the procuring activity.

3.1 1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.2 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-T-21200 for design, construction, and workmanship, except as otherwise specified herein.

3 2.1 Total Weight - The total weight of the equipment, including cables and batteries shall be a minimum consistent with good design and shall not exceed 350 pounds.

3.2 2 Reliability -

3.2.2.1 Operational Stability - The equipment shall operate in a satisfactory manner for a period of at least 1000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use, except that replacement of batteries at intervals of not less than 50 hours of operation will be permitted.

3.2.2.2 Operating Life - The equipment shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to deterioration during the life of the equipment and the rate of deterioration of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.2.3 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200

3.2 4 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200

3 2 5 Interference Control - The generation of radio frequency interference and the susceptibility of the equipment to radio and audio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2 6 Maintenance Provisions - Provisions for maintenance shall be as specified in Specification MIL-T-21200.

3.2 7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307

3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics for standard operation of the equipment.

Temperature	Room ambient (25° ± 15°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	12.5 V DC maximum (low noise battery source)

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3.2.9 Service Conditions - The test equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 3 equipment except as modified herein.

3.2 9.1 Temperature - Without any warm-up period, the equipment shall operate satisfactorily under any possible combination of the ambient conditions specified in Specification MIL-T-21200

3 2.9.2 Shock - The equipment shall withstand a shock of 5g when subjected to impact as specified in Specification MIL-T-21200 except that the Shield, Test, shall not be subjected to any shock

3 2 9 3 Vibration - The test equipment shall not suffer damage or subsequently fail to provide specified performance when subjected to continuous vibration along each of the three mutually perpendicular axes within the frequency range and amplitude as follows, except that the Shield, Test, shall be tested while supported, properly seated on the Dolly, Test Equipment, during the test

Frequency Range	Double Amplitude
5 to 15 cps	0 060 ± 0.006
16 to 25 cps	0.040 ± 0 004

3 2 10 Warm-Up Time - No warm-up time shall be required for the operation of the test equipment described herein Warm-up time of the Magnetic Anomaly Detection equipment under test shall be determined by the particular Magnetic Anomaly Detection equipment

3 2 11 Primary Input Power Requirements - The test equipment shall be capable of satisfactory operation when supplied input power having characteristics as defined in Specification MIL-T-21200 The power required shall be of the following type and shall not exceed the specified amount Batteries shall conform to the requirements of Specification MIL-B-18

(1) DC Power, battery source, 12 5 V DC maximum 0 010 A.

3.3 Performance - The performance of the equipment shall be as set forth below Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions When reduced performance requirements for service conditions are acceptable, acceptable variations from the performance under the standard conditions will be specified

3 3 1 Functional Performance - The Test Set Group, when used in conjunction with the AN/ASM-3 or AN/ASM-6 Test Sets, shall be used for maintenance, repair, and adjustment of the AN/ASQ-8 and AN ASQ-10 Detecting Sets

3 3 2 Prime Equipment Serviced - The Test Set Group shall provide for mounting and magnetically shielding the Detecting Head DT-37(*)/ASQ-8 or DT-239/ASQ-10A. It shall also provide a uniform noise free, controllable, simulated earth's field in the area of the detecting magnetometer of the Detecting Head under test and provide for interconnecting the Detecting Head under test with Cable Assembly, CX-2512/U (W1019-1). of Detecting Set Test Set, AN/ASM-3, or W-7 of Magnetic Detector Test Set. AN/ASM-6 as applicable.

3.3 3 Simulated Earth's Field Characteristics - The simulated earth's field within the Shield, Test, MX-6797(*)/ASM-3, of the Test Set Group shall be noise free and uniform in the area of the detecting magnetometer of the Detecting Head under test The intensity of the simulated earth's field at the detecting magnetometer shall be variable in both directions within the range of 20 000 to 75,000 gamma

3.4 Detail Requirements -

3 4.1 Shield, Test, MX-6797/ASM-3 - This unit shall meet the following requirements

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3.4 1.1 Function - The Shield, Test, shall completely enclose the Magnetic Detector Head, DT-37(*)/ASQ-8, or DT-239/ASQ-10A, when mounted in the lower portion of the inner mounting cradle of Detecting Head Test Stand, MT-1321/ASM-3, and the Magnetic Simulator, SM-417/ASM-3, and shield them from the effects of the Earth's magnetic field and all other fixed and variable magnetic fields to the degree specified herein. It normally rests on the Dolly, Test Equipment, V-341(*)/ASM-3, at all times, when in use, storage, or being transported

3.4 1 2 Form Factor - The Shield, Test, shall not exceed 24 5/8 inches in diameter by 54 inches high. This does not include the electrical connectors and connector mounting box, which shall not extend more than 4 inches beyond the lower periphery of the Shield, Test.

3 4 1 3 Weight - The weight of the Shield, Test, shall not exceed 260 pounds.

3 4.1 4 Contents of Unit - The Shield, Test, shall contain the following assemblies, subassemblies, and circuits

(1) Barrel Assemblies - Four barrel assemblies, secured one within the other, manufactured from a nickel iron alloy having high magnetic permeability at low inductions, shall form a part of the Shield, Test The innermost barrel assembly shall be sufficiently large to accept the complete Detector Head, DT-37(*)/ASQ-8, or DT-239/ASQ-10A, and the Simulator, Magnetic, SM-417/ASM-3, when mounted in the lower portion of the inner mounting cradle of Detecting Head Test Stand, MT-1321/ASM-3. Each barrel assembly shall be uniformly spaced approximately one inch in all directions from the adjacent barrel assembly Each barrel assembly shall be electrically and magnetically insulated from all other barrel assemblies All assembly hardware shall be of non-magnetic material

(2) Lid Assemblies - Four dome shaped lid assemblies, each to fit one barrel assembly and manufactured from the same material as the barrel assemblies, shall form a part of the Shield, Test. Each lid assembly, when installed, shall be uniformly spaced approximately one inch from each adjacent lid assembly and shall overlap the associated barrel assembly a minimum of 1 1/4 inches Cushioning shall be provided to protect the lid assemblies from mechanical shock when removed from the Shield Test, and nested in an inverted position

(3) Circuits - Circuits shall be provided for the following functions

(a) Extension of Operating Circuits of the Detector Head, DT-37(*)/ASQ-8, or DT-239/ASQ-10A.

(b) Extension of Operating Circuits of the Simulator, Magnetic.

3 4.1.5 Electrical Connections - Connections to external circuits shall be provided as follows

Reference Designation	Connector Type	Function
2J1	MS3102R14S-2P	Magnetic Simulator Power from Simulator Control
2J2	MS3102R24-28P	Detector Head Control from Amplifier Test Set, TS-798/ASM-3, or Magnetic Detector Test Set, TS-1193/ASM-6
*2P1	MS3106R14S-2S	Magnetic Simulator Power to Magnetic Simulator

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Reference Designation	Connector Type	Function
*2P2	MS3106R24-28S	Detector Head Control to Detector Head, DT-37(*)/ASQ-8, or DT-239/ASQ-10A

*Modified to remove magnetic materials

2J1 Pin	Purpose	2P1 Pin
A	Spare Lead	A
B	Simulator Power	B
C	Simulator Power	C
D	Shield for A, B and C Leads	D

2J2 Pin	Purpose	2P2 Pin
A	Detector Drive	A
B	Detector Drive	B
C	Detector Drive	C
D	Shield for Detector Drive Leads A, B and C	D
E	Ground	E
F	Shield	F
G	Open	G
H	Open	H
J	Inner Drive	J
K	Inner Drive	K
L	Inner Drive	L
M	Outer Drive	M
N	Outer Drive	N
P	Outer Drive	P
Q	Ground Shield	Q
R	Inner Variable Motor	R
S	Shield	S
T	Outer Variable Motor	T
U	Third Variable Motor Shield	U
V	Third Variable Motor	V
W	Constant Field	W
X	Constant Field	X
Y	Third Input Signal	Y
Z	Third Input Signal	Z

Wiring shall be in accordance with MCAS Cherry Point Drawing No C1-1323

3 4 1 6 Shielding Properties - The magnetic attenuation provided by the Shield Test, shall be the maximum obtainable using proper annealing procedures for the material used. The attenuation shall be not less than 65 db when subjected to a sinusoidal magnetic field of 1,600 gamma, peak to peak, when alternating at a frequency of 2 cycles per second or less, where db = 20 log (attenuation ratio).

3.4.2 Shield, Test, MX-6797A/ASM-3 - This unit shall meet the requirements for Shield, Test, MX-6797/ASM-3, except as follows:

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3.4.2.1 Shielding Properties - The magnetic attenuation provided by the Shield, Test, shall be the maximum obtainable using proper annealing procedures for the material used. The attenuation shall be not less than 95 db when subjected to a sinusoidal magnetic field of 8000 gamma, peak to peak, when alternating at a frequency of 2 cycles per second or less, where: db = 20 log (attenuation ratio).

3.4.3 Dolly, Test Equipment, V-341/ASM-3 - This unit shall meet the following requirements

3.4.3.1 Function - The Dolly, Test Equipment, shall provide cushioned support for the Shield, Test, MX-6797(*)/ASM-3, at all times; when in use, idle, or being transported. It shall also provide a mounting plate for the Control, Simulator, C-6553/ASM-3. The Dolly, Test Equipment, shall be provided with three legs having cushioned feet which shall support the Shield, Test, in a stable, upright position for use or storage. The Dolly, Test Equipment, shall be provided with pneumatic tired wheels which shall contact the floor when it is tilted providing ready mobility. A handle shall be provided to guide and control the Dolly, Test Equipment, at such times that the unit is being moved. A device shall be incorporated or supplied to secure the Shield, Test, in position on the Dolly, Test Equipment, at all times.

3.4.3.2 Form Factor - The Dolly, Test Equipment, shall not exceed 40 inches in length, 31 1/2 inches in width, and 46 inches in height.

3.4.3.3 Weight - The weight of the Dolly, Test Equipment, shall not exceed 58 pounds.

3.4.4 Dolly, Test Equipment, V-341A/ASM-3 - This unit shall meet the requirements for Dolly, Test Equipment, V-341/ASM-3, except as follows:

3.4.4.1 Form Factor - The Dolly, Test Equipment, shall not exceed 40 inches in length, 27 inches in width, and 46 inches in height.

3.4.5 Simulator, Magnetic, SM-417/ASM-3 - This unit shall meet the following requirements

3.4.5.1 Function - The Simulator, Magnetic, shall mount on the lower portion of the inner mounting cradle of Detecting Head Test Stand, MT-1321/ASM-3, utilizing the mounting holes provided for the installation of the upper portion of the inner mounting cradle of the Detecting Head Test Stand, MT-1321/ASM-3. When in use, the Simulator, Magnetic, and Detecting Head under test, both mounted on the lower portion of the inner mounting cradle of Detecting Head Test Stand, MT-1321/ASM-3, shall rest inside the Shield, Test, MX-6797(*)/ASM-3. Four adjustable braces shall be provided on the Simulator, Magnetic, to position the Simulator, Magnetic, and the Detecting Head under test in the center of the Shield, Test. The magnetic field produced by the Simulator, Magnetic, shall be uniform in the area of the detecting magnetometer of the Detecting Head under test.

3.4.5.2 Form Factor - The Simulator, Magnetic, shall not exceed 15 3/4 inches in length, 11 inches in width, and 23 inches in height. This does not include the four braces which shall be adjustable over a diametral range which will permit locking the Simulator, Magnetic, in position within the Shield, Test.

3.4.5.3 Weight - The weight of the Simulator, Magnetic, shall not exceed 12 pounds.

3.4.5.4 Contents of Unit - The Simulator, Magnetic, shall contain the following assemblies, subassemblies, and circuits:

(1) Coil Assemblies - Two coil assemblies shall form a part of the Simulator, Magnetic. They shall be connected in series, aiding, and shall encompass a common axis. The coil assemblies, when installed, shall produce, along the coil axis and midway between the two coils, a magnetic field of uniform density and 20,000 to 75,000 gamma intensity through a maximum current range of 2.0 to 10.0 milliamperes.

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(2) Stand Assembly - The Stand Assembly shall so support the Coil Assemblies that when in use the center of the detecting magnetometer of the Detecting Head under test is on the axis of the Coil Assemblies, midway between the Coil Assemblies, and also such that the Detecting Head under test is allowed complete freedom of movement

(3) Braces - Four easily adjustable Braces shall be provided to rigidly position the Stand Assembly within the Shield, Test, MX-6797(*)/ASM-3, by bearing against the inner Barrel Assembly of the Shield, Test That portion of the brace which bears against the Shield Test, shall be of sufficient size and resiliency to preclude damage to the Shield, Test

3.4.5.5 Electrical Connections - Connections to external circuits shall be provided as follows

Reference Designation	Receptacle Type	Function
3J1	MS3102R14S-2P	Magnetic Simulator Power from Simulator Control through the Shield, Test

3J1 Pin	Purpose
A	No Connection
B	Simulator Power
C	Simulator Power
D	Ground

3.4.5.6 Construction - The Simulator Magnetic, shall be entirely constructed of non-magnetic materials

3.4.6 Control, Simulator C-6553/ASM-3 - This unit shall meet the following requirements.

3.4.6.1 Function - The Control, Simulator shall mount on a mounting plate provided on the Dolly, Test Equipment V-341(*)/ASM-3 and shall contain all the switches, controls, and power supply required for the operation and control of the Simulator, Magnetic, SM-417/ASM-3.

3.4.6.2 Form Factor - The Control, Simulator, shall not exceed 8 inches in length, 6 inches in width, and 6 inches in height

3.4.6.3 Weight - The weight of the Control Simulator, shall not exceed five pounds including the power supply

3.4.6.4 Contents of Unit - The Control Simulator, shall contain the following assemblies, subassemblies, and circuits

(1) Power Supply - A D C. power supply shall form a part of the Control Simulator The power supply shall provide a maximum of 12.5 volts D C for the operation of the Simulator, Magnetic, SM-417/ASM-3 from a battery source conforming to MIL-B-18, or other source being equally noise free and stable. The output current of the power supply shall be controllable in direction and in magnitude from 2.0 milliamperes to 10 milliamperes into a 400 ohm load in a minimum of ten steps. Resistors and capacitors shall conform to Specifications MIL-R-11 and MIL-C-25, respectively. Electrical current magnitude controls shall be of the non-interrupting type (continuously variable or shorting type in steps).

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3.4.6.5 Controls - The following controls shall be provided on the Control, Simulator:

(1) Toggle switch with ON, center OFF and ON positions - This switch shall control direction of the current to the Simulator, Magnetic, and apply (or remove) D.C. power to (or from) the Simulator, Magnetic.

(2) Rotary control, INCREASE FIELD - This control shall vary the magnitude of the current delivered to the Simulator, Magnetic, increasing the current as the switch is turned clockwise.

3.4.6.6 Electrical Connections - Connections to external circuits shall be provided as follows

Reference Designation	Receptacle Type	Function
1J1	MS3102R14S-2P	Power and Control for Simulator, Magnetic

1J1 Pin	Purpose
A	No Connection
B	Simulator Power
C	Simulator Power
D	Ground

3.4.7 Cable Assembly, Special Purpose, Electrical, W-1 of OA-7366/ASM-3 - A cable assembly shall be provided with a plug, W1P1, on one end, and W1P2, on the other end, and be of adequate length to interconnect the Control, Simulator, C-6553/ASM-3, to the Shield, Test, MX-6797(*)/ASM-3 as follows

Reference Designation	Connector Type
W1P1	MS3106R14S-2S
W1P2	MS3106R14S-2S

W1P1 Pin	Purpose	W1P2 Pin
A	Spare Lead	A
B	Simulator Power	B
C	Simulator Power	C
D	Shield for Leads A, B and C	D

3.5 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment

Type Designation	Nomenclature
AN/ASM-3	Test Set, Detecting Set
AN/ASM-6	Test Set, Magnetic Detector

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract.

(1) Contractor Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor Demonstration Tests - Contractor demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5 The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory

4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected during conduction of these tests to the procuring agency for review and approval

4.2.2 Service Approval Tests - At the completion of contractor demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design test data specified in Specification MIL-T-21200 This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental testing in accordance with Specification MIL-T-5422, modifying the test limits and the applicable tests to those specified herein Interference tests shall be conducted in accordance with Specification MIL-I-6181

4.2.4 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests No production equipment shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance The preproduction sample shall not be considered as one of the equipments under the contract, however, it may be reworked by the contractor and submitted for acceptance as a production equipment

4.2.5 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection, or if already accepted by the government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring agency.

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4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent by the contractor at his expense to a designated government laboratory for tests. This equipment shall be selected by the government's inspector after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test

4 3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the approved preproduction sample in design, construction. workmanship, performance and quality, and that it is capable of meeting all applicable requirements

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity

4 3.3 Reconditioning of Production Test Sample - On completion of the production sample test, the equipment shall be reworked by the contractor by replacing all wear or damaged items After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall be responsible for accomplishing the acceptance tests. The contractor shall furnish all samples and shall accomplish the sampling tests along with all other tests All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the services of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or the laboratory, as applicable Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4 4 1 Individual Tests - Each equipment submitted for acceptance by the procuring activity shall be subjected to an individual test. The portions of this test shall be adequate to determine compliance with the requirements of material workmanship operational adequacy and reliability As a minimum each equipment accepted shall have passed the following tests

(1) Examination of Product

(2) Operational Test

4.4.1 1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation. The shielding efficiency of each Shield, Test, shall be certified to be not less than the required value specified in paragraph 3.4.1.6 or 3.4.2.1 above, as applicable, and a decal bearing the actual attenuation value (to the nearest decibel) shall be affixed to each Shield, Test, adjacent to the nameplate. No discernable magnetic noise shall be introduced into the equipment under test (when adjusted to its maximum sensitivity) by the Magnetic Detector Head Test Set Group, OA-7366(*)/ASM-3.

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4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1
Next 25	1
Next 25	1
Next 50	1 and 1 for each additional 100 or fraction thereof

4 4.2 1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met

(2) Operational test at certain environmental conditions The conditions may vary for each equipment tested and should be based on results of the preproduction, production, and special tests

(3) A manufacturing reliability test wherein each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure A failure shall be defined as anything which causes malfunctioning of the equipment Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment

Temperature	Ambient Room
Humidity	Ambient Room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3g's. (This test shall not apply to the Shield, Test)

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used During the 6-hour period of operation following the 10-minute vibration period the equipment shall be mechanically cycled periodically through its various phases of operation Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

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4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows: except that when a Production Sample is requested and delivered to a government laboratory for test (See paragraph 4.3), the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity).

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Tests procedures previously approved for the preproduction and sampling tests shall be used where applicable When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Tests (60 minutes cycling in each of 3 planes - except the Shield, Test)

(3) Humidity Tests

4 4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or a design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval The right is reserved by the procuring activity or the Government Inspector to modify or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity The test procedure for the reliability tests shall be prepared by the contractor and approved by the procuring activity.

4.6 Presubmission Testing - No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order, or in the absence of such requirement, Level A, Method III of MIL-E-17555, shall be used.

6. NOTES

6.1 Intended Use - The Test Set Group, Magnetic Detector Head, OA-7366(*)/ASM-3, shall be used for the maintenance, repair, and adjustment of Detecting Sets, Magnetic, AN/ASQ-8 and AN/ASQ-10. It shall be designed to be used in conjunction with Test Set, Detecting Set, AN/ASM-3, and Test Set, Magnetic Detector, AN/ASM-6, to provide a space for the magnetic detector head which will be free of magnetic disturbances and provide a controlled simulated earth's field within the area of the detector head.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specifications.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified in Specification MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 Type Designation - The parentheses (*) when used in the type designation shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with paragraph 3.2.7. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patended invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-T-81308(WP) Test Set Group, Magnetic Detector Head, OA-7366/ASM-3, OA-7366A/ASM-3 and OA-7366B/ASM-3

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Washington, D. C. 20360

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